

Systems of linear equations

In general we will study **systems of linear equations**.

This means we consider a *list* of linear equations all of which use the same unknowns. For example consider the following equations in the three variables x_1, x_2 and x_3 :

$$\begin{cases} x_1 - x_2 = 15 \\ x_1 + 2x_2 - x_3 = 25 \\ x_1 - x_3 = 5 \end{cases} \quad \text{list the variables in advance}$$

This will be systematic if we put every variable in:

$$\begin{cases} x_1 - x_2 + 0x_3 = 15 \\ x_1 + 5x_2 - x_3 = 25 \\ x_1 + 0x_2 - x_3 = 5 \end{cases}$$

Ordering of the equations:

So we can be precise later:

When we refer to a “system of linear equations” we are assuming we are **listing the equations in a particular order**.

For example:

$$\begin{cases} x_1 - x_2 + 0x_3 = 15 \\ x_1 + 5x_2 - x_3 = 25 \\ x_1 + 0x_2 - x_3 = 5 \end{cases} \quad \begin{array}{l} \text{Equation \#1.} \\ \text{Equation \#2.} \\ \text{Equation \#3.} \end{array}$$

(So note that we have an ordering of *both* the variables and also the list of equations. Usually this choice is just in the background and we don't say it explicitly.)

A system of linear equations:

$$\begin{cases} x_1 - x_2 + 0x_3 = 15 \\ x_1 + 5x_2 - x_3 = 25 \\ x_1 + 0x_2 - x_3 = 5 \end{cases}.$$

A **solution** is a choice of a value for each variable which makes ALL the equations true, *simultaneously*.

For example, you can check:

- $x_1 = 19, x_2 = 4, x_3 = 14$ *IS* a solution because it makes all three equations true simultaneously.
- $x_1 = -1, x_2 = -16, x_3 = 0$ is *NOT* a solution (even though it solves the first equation, it doesn't solve the other two).

Example

Consider a **cubic curve** with equation

$$y = a + bx + cx^2 + dx^3,$$

where a, b, c, d are real constants.

What system of equations do these constants a, b, c, d have to satisfy so that this curve passes through the points

$$(0, 10), (1, 7), (3, -11) \text{ and } (4, -14)?$$

Solution

Every point the curve is supposed to go through gives us a different linear equation.

For example, substituting $x = 3$ and $y = -11$ into the equation leads to:

$$-11 = a + b(3) + c(3)^2 + d(3)^3.$$

Overall, the four points lead to a system of 4 linear equations:

$$\begin{cases} a & = 10 \\ a + b + c + d & = 7 \\ a + 3b + 9c + 27d & = -11 \\ a + 4b + 16c + 64d & = -14 \end{cases}$$

Earlier we defined “solution set” for just a single linear equation. Of course this generalizes in an obvious way to a system of linear equations.

Video 2.1.2

Definition of the “solution set of a system of linear equations”.

Fix a system of linear equations in n variables and assume the variables of the equation are in some order. (For example maybe the variables are $x_1, \dots, x_n$ in the obvious order).

The **solution set** of the system of linear equations is the subset of $\mathbb{R}^n$

$$S \subset \mathbb{R}^n$$

consisting of all n -tuples that correspond to solutions of the system of equations.

Comment:

Some systems of linear equations have **no** solutions at all. In other words, their set of solutions is empty. In this case we say that their set of solutions is the empty set $\emptyset \subset \mathbb{R}^n$.

empty set notation

For example:

$$\begin{cases} x + y = 2 \\ 2x + 2y = 6 \end{cases}$$

If you take any pair of numbers (x, y) which satisfies the first equation and substitute it into the second equation you get:

$$2x + 2y = 2(x + y) = 2 * 2 = 4 \neq 6.$$

So it is impossible to satisfy both equations simultaneously.

Inconsistency

Definition

A system of linear equations that has no solutions is said to be **inconsistent**.

On the other hand, if a system has at least one solution, then it is said to be **consistent**.

How many solutions can a system have?

Later on we will prove an interesting fact.

There are *only three possibilities* for a linear system:

- It has no solutions. (In other words, it is inconsistent.)
- It has exactly one solution.
- It has infinitely many solutions.

} consistent

For example: There does not exist a linear system with exactly 7 solutions.

Either have 1 solⁿ or infinite solⁿ.

over $\mathbb{R}$ numbers
↓
will learn more if you do pure mathematics

Video 2.1.3

Example: A system of two equations in two variables.

Consider the following system

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

For the discussion, assume that at least one of a_1 and b_1 is non-zero, and similarly that at least one of a_2 and b_2 is non-zero. (If they are both zero we are in some special case. Later we'll include the special cases.)

Question:

What are the possibilities for the number of solutions of a system of this type?

We haven't discussed the algebraic details yet, but anyway maybe we can understand the possibilities geometrically?

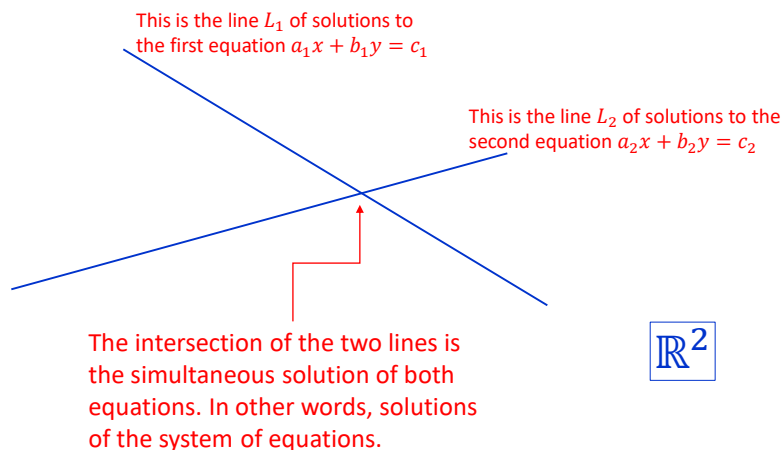
To understand the set of solutions to the given system

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases} \text{ note the following sequence of observations:}$$

1. Solutions of the first equation form some line L_1 in the plane $\mathbb{R}^2$.
2. Solutions of the second equation also form some line L_2 in the plane $\mathbb{R}^2$.
3. What we want is solutions of the total system - in other words, solutions which satisfy both equations *simultaneously*.
4. These will be the points that lie in the *INTERSECTION* of the two lines $L_1 \cap L_2$.
5. So we ask: *What are the possible ways two lines can intersect in the plane?*

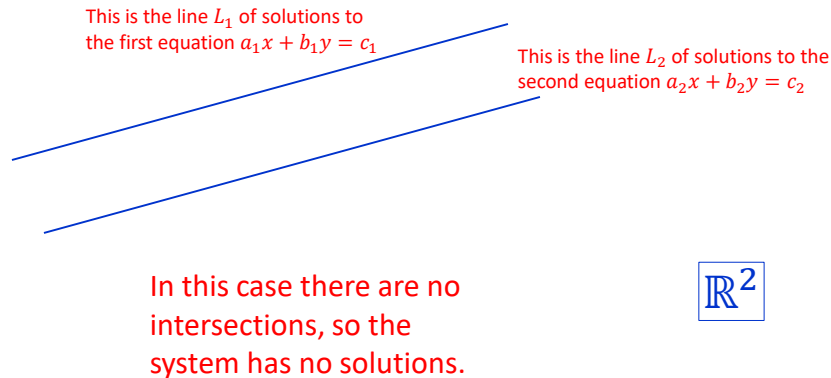
There are exactly three geometric possibilities:

Possibility #1: The lines *intersect once*, corresponding to *exactly one solution* to the system.



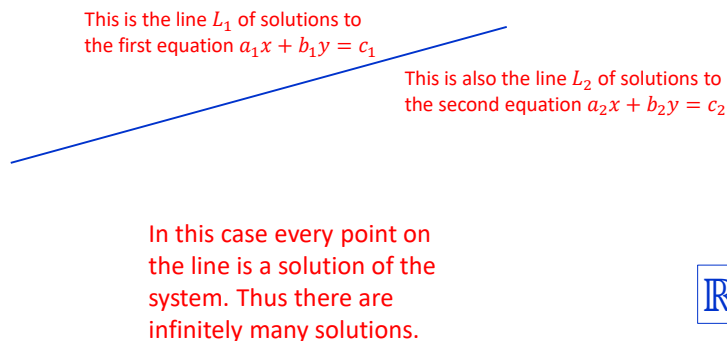
There are exactly three geometric possibilities:

Possibility #2: The lines **never intersect** (so they must be parallel), corresponding to **no solutions**, or in other words an inconsistent system.



There are exactly three geometric possibilities:

Possibility #3: The **two lines are equal** $L_1 = L_2$ so the whole line lies in the intersection, so there are **infinitely many solutions**.



As soon as you have 2 solutions, you have an infinite amount of solutions.

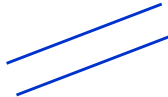
Summary.

We studied the following system geometrically

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

We found that as expected there are exactly three possibilities.

1. No solutions.



2. Exactly one solution.



2. Infinitely many solutions.



Example: The case of 3 equations in 3 variables

Video 2.1.4

Consider the following system of three linear equations in three variables:

$$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases}$$

The set of solutions for a linear equation with three variables is a plane in 3-d space.

We want to use geometric intuition to understand the possibilities for the set of solutions. (For the discussion assume there are no equations where every coefficient on the LHS is 0.)

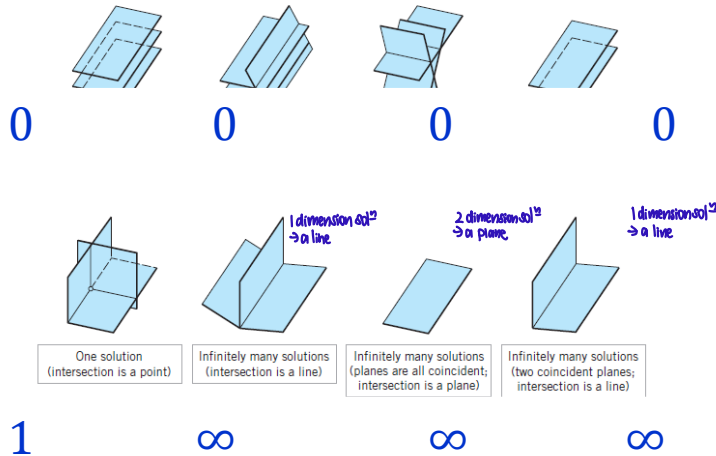
The set of solutions for each equation separately form a plane.

The set of solutions of the whole system are the points that lie in *all three planes*.

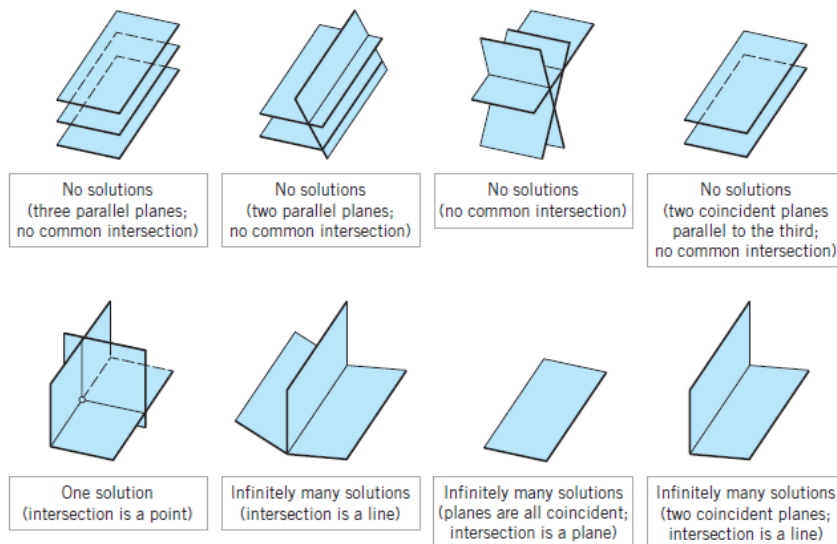
What are the possibilities?

Here is a list of the geometric possibilities. In each case, how many solutions are there?

*There are only 3 possible types of solution: 0, 1 or infinite



Summary:



Video 2.1.5

The most important thing we have to learn in this course:

How can we compute and understand the complete set of solutions to an arbitrary system of linear equations (with any number of variables and any number of equations)?

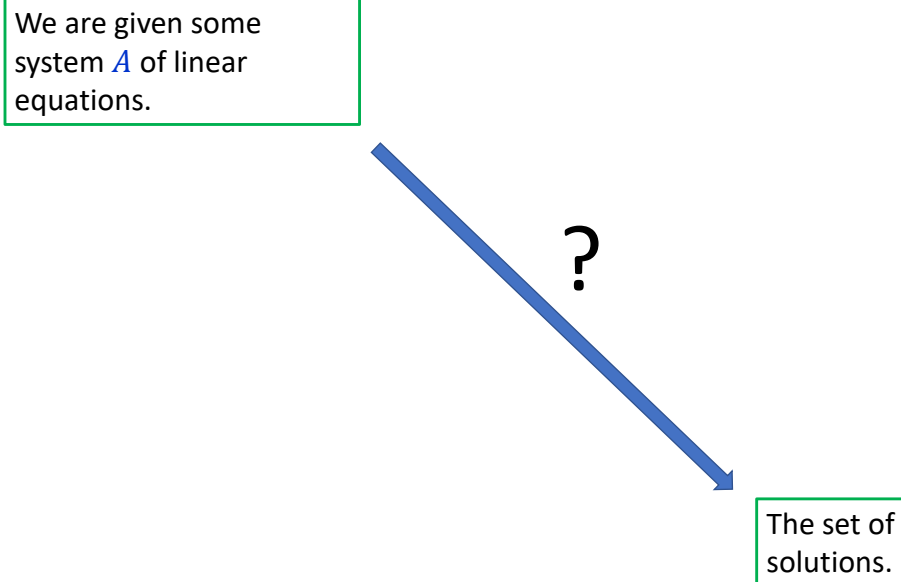
Our main task:

How to determine the solution set of a system of linear equations?

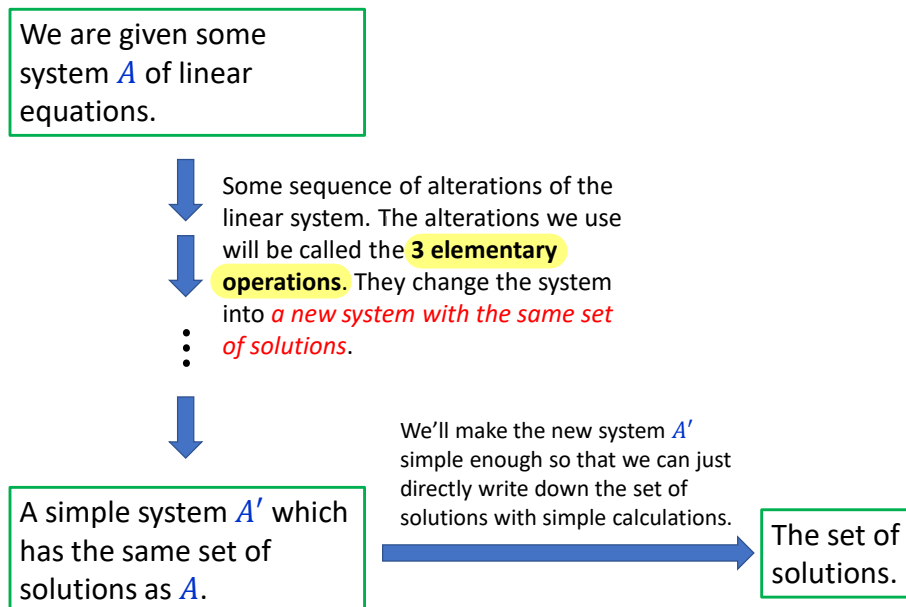
The basic strategy:

- We'll find **3 operations** we can do which will replace a system of equations with another system of equations having the **same solution set** as the original system.
- Then we'll use these operations to **repeatedly simplify** the system of equations until it is in a simple enough form where we can find the solutions directly.

An illustration of the strategy:



An illustration of the strategy:



$$\begin{cases} x_1 + 2x_2 + 3x_3 = 7 \\ -2x_1 + 0x_2 + 10x_3 = -\pi \\ \downarrow \text{Eqn 2} \rightarrow \text{Eqn 2} + 2\text{Eqn 1} \\ \text{means replaced by} \\ x_1 + 2x_2 + 3x_3 = 7 \\ (-2+2)x_1 + (0+4)x_2 + (10+6)x_3 = -\pi + 14 \end{cases}$$

They have the same set of solutions $\rightarrow$ illustrates first elementary operation

The 3 elementary operations.

Video 2.1.6

$$\begin{cases} x_1 + x_2 = 1 \\ -x_1 + 7x_2 = -1000 \\ \downarrow \text{Swap eqn 1 and eqn 2} \\ -x_1 + 7x_2 = -1000 \\ x_1 + x_2 = 1 \end{cases}$$

This illustrates the second elementary operation

$$\begin{cases} x_1 + x_2 = 1 \\ -2x_1 + 1000x_2 = 2020 \\ \downarrow \text{Eqn 2} \rightarrow \frac{1}{2} \text{Eqn 2} \\ x_1 + x_2 = 1 \\ -x_1 + 500x_2 = 1010 \end{cases}$$

This illustrates the third elementary operation

Claim

Consider the following three operations on systems of linear equations. (By "operation" we just mean it is a specific way of altering a system of linear equations into a new system of linear equations.)

1. Replace one of the equations on the list with the equation you get by taking that equation and adding to it a constant multiple of one of the other equations.
2. Keep exactly the same equations but swap the position of two equations on the list.
3. Replace one of the equations on the list by the same equation but after multiplying both sides by the same constant non-zero scalar.

Then: The resulting system of linear equations has exactly the same set of solutions as the system of linear equations you started with.

These three operations are called the **elementary operations**.

Example

$$\begin{cases} x_1 + x_2 + x_3 = -2 \\ -7x_1 + 2x_2 = 0 \\ x_2 - 5x_3 = 1 \end{cases}$$

Operation 3:
Eqn 1 has been multiplied by 7.

$$\begin{cases} 7x_1 + 7x_2 + 7x_3 = -14 \\ -7x_1 + 2x_2 = 0 \\ x_2 - 5x_3 = 1 \end{cases}$$

Operation 1:
Eqn 2 replaced by
Eqn 2 + 1*Eqn 1.

$$\begin{cases} 7x_1 + 7x_2 + 7x_3 = -14 \\ 9x_2 + 7x_3 = -14 \\ x_2 - 5x_3 = 1 \end{cases}$$

Operation 2:
Eqn 2 and 3 have been swapped.

$$\begin{cases} 7x_1 + 7x_2 + 7x_3 = -14 \\ x_2 - 5x_3 = 1 \\ 9x_2 + 7x_3 = -14 \end{cases}$$

These two systems will have the same set of solutions because the second one is obtained from the first by a sequence of elementary operations.

Discussion

It is fairly obvious that elementary operations 2 and 3 result in a new system of linear equations with exactly the same set of solutions.

Example: $\begin{cases} x_1 + x_2 = 3 \\ x_1 - 4x_2 = 0 \end{cases}$ and $\begin{cases} -10x_1 - 10x_2 = -30 \\ x_1 - 4x_2 = 0 \end{cases}$ have the exactly the same set of solutions, because any pair of x_1 and x_2 satisfying $x_1 + x_2 = 3$ will satisfy $(-10)(x_1 + x_2) = (-10)3$, and vice versa.

But for operation 1 this fact is not as obvious.

Video 2.1.7

Proof operation 1 doesn't change the set of solutions.

Consider two linear systems in n variables $x_1, x_2, \dots, x_n$.

(Besides these variables, everything else below is a scalar constant.)

Linear system L_1 :

$$\begin{cases} a_1x_1 + a_2x_2 + \dots + a_nx_n = p \\ b_1x_1 + b_2x_2 + \dots + b_nx_n = q \end{cases}$$

Linear system L_2 :

$$\begin{cases} a_1x_1 + a_2x_2 + \dots + a_nx_n = p \\ (ca_1 + b_1)x_1 + (ca_2 + b_2)x_2 + \dots + (ca_n + b_n)x_n = cp + q \end{cases}$$

Notice that linear system L_2 is obtained by applying elementary operation 1 to L_1 .

(In general the list of equations may be longer but the changes to the proof for that case are trivial. We'll do two equations to simplify.)

Proof (cont.)

Linear system L_1 :

$$\begin{cases} a_1x_1 + a_2x_2 + \cdots + a_nx_n = p \\ b_1x_1 + b_2x_2 + \cdots + b_nx_n = q \end{cases}$$

Linear system L_2 :

$$\begin{cases} a_1x_1 + a_2x_2 + \cdots + a_nx_n = p \\ (ka_1 + b_1)x_1 + (ka_2 + b_2)x_2 + \cdots + (ka_n + b_n)x_n = kp + q \end{cases}$$

Let:

- S_1 denote the set of solutions to L_1 .
- S_2 denote the set of solutions to L_2 .

Our problem: Explain why $S_1 = S_2$.

Proof (cont.)

Our problem: Explain why $S_1 = S_2$.

The way we show that two sets are exactly equal in mathematics is to explain that each of the two sets is contained in the other side:

1. $S_1 \subset S_2$
2. $S_2 \subset S_1$

So we must explain why every solution of L_1 is also a solution of L_2 ...

... and we must ALSO explain why every solution of L_2 is a solution of L_1 .

Proof (cont.)

Part 1: Why is a solution of L_1 also a solution of L_2 ?

In this part we assume: we have a solution of linear system L_1 :

$$\begin{cases} a_1x_1 + a_2x_2 + \cdots + a_nx_n = p & (1) \\ b_1x_1 + b_2x_2 + \cdots + b_nx_n = q & (2) \end{cases}$$

Must prove linear system L_2 is satisfied also:

$$\begin{cases} a_1x_1 + a_2x_2 + \cdots + a_nx_n = p \\ (ka_1 + b_1)x_1 + (ka_2 + b_2)x_2 + \cdots + (ka_n + b_n)x_n = kp + q \end{cases}$$

So assume that $(x_1, \dots, x_n)$ is a solution of L_1 . Why does it solve L_2 as well? The first equation is exactly the same in both systems so we only need to focus on the second equation of L_2 :

$$\begin{aligned} LHS &= (ka_1 + b_1)x_1 + (ka_2 + b_2)x_2 + \cdots + (ka_n + b_n)x_n \\ &= k(a_1x_1 + \cdots + a_nx_n) + (b_1x_1 + \cdots + b_nx_n) \\ &= k * p + q \quad \leftarrow \text{Using (1) and (2).} \\ &= RHS. \end{aligned}$$

Proof (cont.)

Part 2: Why is a solution of L_2 also a solution of L_1 ?

We assume linear system L_2 is solved:

$$\begin{cases} a_1x_1 + a_2x_2 + \cdots + a_nx_n = p & (1) \\ (ka_1 + b_1)x_1 + (ka_2 + b_2)x_2 + \cdots + (ka_n + b_n)x_n = kp + q & (2) \end{cases}$$

Must show linear system L_1 holds also:

$$\begin{cases} a_1x_1 + a_2x_2 + \cdots + a_nx_n = p \\ b_1x_1 + b_2x_2 + \cdots + b_nx_n = q \end{cases}$$

Assume that $(x_1, \dots, x_n)$ is a solution of L_2 . Why does it solve L_1 as well? Again, we only need to focus on the second equation of L_1 :

$$\begin{aligned} LHS &= b_1x_1 + \cdots + b_nx_n \\ &= (b_1x_1 + \cdots + b_nx_n) + k(a_1x_1 + \cdots + a_nx_n) - k(a_1x_1 + \cdots + a_nx_n) \\ &= ((b_1 + ka_1)x_1 + \cdots + (b_n + ka_n)x_n) - k(a_1x_1 + \cdots + a_nx_n) \\ &= (kp + q) - k(p) \quad \leftarrow \text{Using (1) and (2).} \\ &= RHS \end{aligned}$$

□

Video 2.1.8

The next thing we'll do is learn a way to present our equations in an efficient form that will be easy to do computations with.

The augmented matrix

Consider a system of m linear equations in n variables as follows. Note that:

1. The variables are ordered, as usual: $x_1, \dots, x_n$.
2. Also the equations are ordered as well.

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n & = & b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n & = & b_2 \\ \vdots & & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n & = & b_m \end{cases}$$

Definition (Augmented matrix.)

Consider some linear system consisting of an ordered list of m linear equations in n variables:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n & = & b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n & = & b_2 \\ \vdots & & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n & = & b_m \end{cases}$$

The **augmented matrix** corresponding to this system is the following $m \times (n + 1)$ matrix:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right]$$

Note: if you order the variables, or the equations in the system in a different way, then this matrix will be rearranged.

Comment:

$$a_{ij}$$

When we refer to positions in a matrix, the first index always refers to the row number...

... and the second index always refers to the column number.

Example

Consider the system of linear equations:

$$\begin{cases} x_1 + x_2 + 2x_3 &= 9 \\ 2x_1 + 4x_2 - 3x_3 &= 1. \\ 3x_1 + 6x_2 - 5x_3 &= 0 \end{cases}$$

The corresponding augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array} \right].$$

Exercise

Video 2.1.9

Consider the following system of linear equations, with the variables ordered a, b, c, d, e and the equations ordered as follows:

$$\begin{cases} d - e = 1 \\ b - d = a - c \\ 10 = e - d + c - b + a \\ a + b + c + d + e = 1 \end{cases} \quad \begin{array}{l} d - e = 1 \\ -a + b + c - d = 0 \\ -a + b - c + d - e = -10 \\ a + b + c + d + e = 1 \end{array}$$

What is a corresponding augmented matrix?

$$\left[\begin{array}{ccccc|c} 0 & 0 & 0 & 1 & -1 & 1 \\ -1 & 1 & 1 & -1 & 0 & 0 \\ -1 & 1 & -1 & 1 & -1 & -10 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{array} \right]$$

You will get a different but correct answer if you swap the LHS and RHS of any equation during this. The row will be multiplied by -1

The elementary row operations.

Video 2.1.10

Earlier we met operations on systems of linear equations we call the **3 elementary operations**. These operations replace a system of equations with a new system having the same set of solutions.

The corresponding operations on the augmented matrices which organize systems of linear equations will be called **elementary row operations**.

Let's do examples of the three different kinds of elementary operation to work out what the elementary row operations are, and then summarise with a definition.

Elementary operation #1:

#1: Add a multiple of one equation to another equation.

$$\begin{array}{l}
 x_1 + 3x_2 - 7x_3 + 28x_4 = 3 \\
 -x_1 + 10x_2 + x_3 - 20x_4 = 0 \\
 x_1 + x_2 - x_3 - x_4 = 1
 \end{array}
 \quad \rightarrow \quad
 \left[\begin{array}{cccc|c}
 1 & 3 & -7 & 28 & 3 \\
 -1 & 10 & 1 & -20 & 0 \\
 1 & 1 & -1 & -1 & 1
 \end{array} \right]
 \begin{array}{l}
 \rightarrow \text{multiply by } -1 \\
 \downarrow \\
 \text{add to row 3}
 \end{array}$$

$$\begin{array}{l}
 \downarrow \text{ Add } (-1) \text{ times equation 1} \\
 \text{to equation 3}
 \end{array}
 \quad
 \begin{array}{l}
 \downarrow \text{ Add } (-1) \text{ times row 1} \\
 \text{to row 3.}
 \end{array}$$

$$\begin{array}{l}
 x_1 + 3x_2 - 7x_3 + 28x_4 = 3 \\
 -x_1 + 10x_2 + x_3 - 20x_4 = 0 \\
 0x_1 - 2x_2 + 6x_3 - 29x_4 = -2
 \end{array}
 \quad \rightarrow \quad
 \left[\begin{array}{cccc|c}
 1 & 3 & -7 & 28 & 3 \\
 -1 & 10 & 1 & -20 & 0 \\
 0 & -2 & 6 & -29 & -2
 \end{array} \right]$$

Elementary operation #2:

#2: Interchange two equations

$$\begin{array}{l}
 x_1 + 3x_2 - 7x_3 + 28x_4 = 3 \\
 -x_1 + 10x_2 + x_3 - 20x_4 = 0 \\
 x_1 + x_2 - x_3 - x_4 = 1
 \end{array}
 \quad \rightarrow \quad
 \left[\begin{array}{cccc|c}
 1 & 3 & -7 & 28 & 3 \\
 -1 & 10 & 1 & -20 & 0 \\
 1 & 1 & -1 & -1 & 1
 \end{array} \right]$$

$$\begin{array}{l}
 \downarrow \text{ Swap equation two} \\
 \text{and three}
 \end{array}
 \quad
 \begin{array}{l}
 \downarrow \text{ Swap rows two} \\
 \text{and three}
 \end{array}$$

$$\begin{array}{l}
 x_1 + 3x_2 - 7x_3 + 28x_4 = 3 \\
 x_1 + x_2 - x_3 - x_4 = 1 \\
 -x_1 + 10x_2 + x_3 - 20x_4 = 0
 \end{array}
 \quad \rightarrow \quad
 \left[\begin{array}{cccc|c}
 1 & 3 & -7 & 28 & 3 \\
 1 & 1 & -1 & -1 & 1 \\
 -1 & 10 & 1 & -20 & 0
 \end{array} \right]$$

Elementary operation #3:

#3: Multiply an equation by a **nonzero constant**.

$$\begin{array}{l}
 x_1 + 3x_2 - 7x_3 + 28x_4 = 3 \\
 -x_1 + 10x_2 + x_3 - 20x_4 = 0 \\
 x_1 + x_2 - x_3 - x_4 = 1
 \end{array}
 \quad \longrightarrow \quad
 \left[\begin{array}{cccc|c}
 1 & 3 & -7 & 28 & 3 \\
 -1 & 10 & 1 & -20 & 0 \\
 1 & 1 & -1 & -1 & 1
 \end{array} \right]$$



Multiply the second equation by 5



Multiply the second row of the augmented matrix by 5

$$\begin{array}{l}
 x_1 + 3x_2 - 7x_3 + 28x_4 = 3 \\
 -5x_1 + 50x_2 + 5x_3 - 100x_4 = 0 \\
 x_1 + x_2 - x_3 - x_4 = 1
 \end{array}
 \quad \longrightarrow \quad
 \left[\begin{array}{cccc|c}
 1 & 3 & -7 & 28 & 3 \\
 -5 & 50 & 5 & -100 & 0 \\
 1 & 1 & -1 & -1 & 1
 \end{array} \right]$$

Definition (The elementary row operations.)

Let M and M' be matrices or augmented matrices of the same shape. We say that M' has been obtained from M via an **elementary row operation** if it has been obtained by altering M in one of the following three ways:

Elementary row operation #1:

Adding a multiple of one of the rows to another one of the rows. If row i is replaced by row i plus a constant times row j then we write $R_i \rightarrow R_i + cR_j$ to denote this operation.

Elementary row operation #2:

Swapping two rows. If rows i and j are swapped then we write $R_i \leftrightarrow R_j$ to denote this operation.

Elementary row operation #3:

Multiplying every entry in a row by the same **non-zero scalar**. If row i is replaced by row i multiplied by $c \in \mathbb{R}, c \neq 0$, then we write $R_i \rightarrow cR_i$ to denote this operation.

Examples:

Elementary row operation #3:

$$\begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & -3 \\ 3 & 2 & 1 \end{bmatrix} \xrightarrow{R_2 \rightarrow (-5)R_2} \begin{bmatrix} 1 & 0 & 2 \\ -5 & -5 & 15 \\ 3 & 2 & 1 \end{bmatrix}.$$

Elementary row operation #2:

$$\begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & -3 \\ 3 & 2 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 0 & 2 \\ 3 & 2 & 1 \\ 1 & 1 & -3 \end{bmatrix}.$$

Elementary row operation #1:

$$\begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & -3 \\ 3 & 2 & 1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + (-3)R_2} \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & -3 \\ 0 & -1 & 10 \end{bmatrix}.$$

Caution: Please learn this list of 3 operations *precisely*. There are other simple operations that are sometimes useful but are not on this list (for example, swapping columns). These are **NOT** considered “**elementary row operations**”.

$$\begin{bmatrix} 1 & 2 & ? \\ 4 & 5 & 6 \\ 3 & 7 & 9 \end{bmatrix}$$

↓ Not an elementary row operation
→ will get different answer

$$\begin{bmatrix} 1 & 2 & ? \\ 4 & 5 & 6 \\ 3 & 7 & 9 \end{bmatrix}$$

Video 2.1.11

The general strategy for solving a system of equations.

1. Start with a system of linear equations.
2. Replace it by it's corresponding augmented matrix.
3. Do elementary row operations on the augmented matrix until the matrix is so simple that we can read off the solutions of the corresponding system of equations directly.

Example: Let's solve these equations.

Consider the following system of equations:

$$\begin{cases} x + y + 3z = 0 \\ 2x - 2y + 2z = 4 \\ 3x + 9y = 3 \end{cases}$$

The corresponding augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 2 & -2 & 2 & 4 \\ 3 & 9 & 0 & 3 \end{array} \right]$$

Example: (cont.)

We'll do elementary row operations to simplify the augmented matrix. (We will show how these correspond to elementary operations on the original system of equations at the same time).

The first step will be to replace R_2 by $R_2 - 2R_1$.

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 2 & -2 & 2 & 4 \\ 3 & 9 & 0 & 3 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -4 & -4 & 4 \\ 3 & 9 & 0 & 3 \end{array} \right]$$

1st step is to make this a zero
2nd step is to make this a zero

Example: (cont.)*Augmented matrix:*

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 2 & -2 & 2 & 4 \\ 3 & 9 & 0 & 3 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -4 & -4 & 4 \\ 3 & 9 & 0 & 3 \end{array} \right]$$

Associated linear system:

$$\begin{cases} x + y + 3z = 0 \\ 2x - 2y + 2z = 4 \\ 3x + 9y = 3 \end{cases} \rightarrow \begin{cases} x + y + 3z = 0 \\ -4y - 4z = 4 \\ 3x + 9y = 3 \end{cases}$$

Same set of solutions as the original system.

Example: (cont.)*Augmented matrix:*

$$\dots \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -4 & -4 & 4 \\ 3 & 9 & 0 & 3 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 3R_1} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -4 & -4 & 4 \\ 0 & 6 & -9 & 3 \end{array} \right]$$

3rd step is to make this a zero

Associated linear system:

$$\dots \rightarrow \begin{cases} x + y + 3z = 0 \\ -4y - 4z = 4 \\ 3x + 9y = 3 \end{cases} \rightarrow \begin{cases} x + y + 3z = 0 \\ -4y - 4z = 4 \\ 6y - 9z = 3 \end{cases}$$

Same set of solutions as the original system.

Example: (cont.)*Augmented matrix:*

$$\dots \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -4 & -4 & 4 \\ 0 & 6 & -9 & 3 \end{array} \right] \xrightarrow{R_2 \rightarrow -\frac{1}{4}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 6 & -9 & 3 \end{array} \right]$$

Associated linear system:

$$\dots \rightarrow \begin{cases} x + y + 3z = 0 \\ -4y - 4z = 4 \\ 6y - 9z = 3 \end{cases} \rightarrow \begin{cases} x + y + 3z = 0 \\ y + z = -1 \\ 6y - 9z = 3 \end{cases}$$

Same set of solutions as
the original system.

Example: (cont.)*Augmented matrix:*

$$\dots \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 6 & -9 & 3 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 6R_2} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -15 & 9 \end{array} \right]$$

Associated linear system:

$$\dots \rightarrow \begin{cases} x + y + 3z = 0 \\ y + z = -1 \\ 6y - 9z = 3 \end{cases} \rightarrow \begin{cases} x + y + 3z = 0 \\ y + z = -1 \\ -15z = 9 \end{cases}$$

Same set of solutions as the original system.
But now this set of equations is simple enough to solve directly.

Example: (cont.)

$$\begin{cases} x + y + 3z = 0 \\ 2x - 2y + 2z = 4 \\ 3x + 9y = 3 \end{cases}$$

The system of linear equations we started with.



1. $R_2 \rightarrow R_2 - 2R_1$
2. $R_3 \rightarrow R_3 - 3R_1$
3. $R_2 \rightarrow -\frac{1}{4}R_2$
4. $R_3 \rightarrow R_3 - 6R_2$

$$\begin{cases} x + y + 3z = 0 \\ y + z = -1 \\ -15z = 9 \end{cases}$$

This system is a simple system with the same set of solutions as the original system. All we need to do now is to solve it.

Example: (cont.)

Exercise: Solve this system:
$$\begin{cases} x + y + 3z = 0 \\ y + z = -1 \\ -15z = 9 \end{cases}$$

Solution:

The third equation implies $z = -\frac{3}{5}$.

Then we substitute $z = -\frac{3}{5}$ into the second equation, which implies $y = -\frac{2}{5}$.

Then we substitute $y = -\frac{2}{5}$ and $z = -\frac{3}{5}$ into the original equation which implies $x = \frac{4}{5}$.

So there is exactly one solution. $(\frac{4}{5}, -\frac{2}{5}, -\frac{3}{5})$



Example: (cont.)

The way we solve this simplified system has a name. It is called **back-substitution**.

Example: (Summary)

$$\begin{cases} x + y + 3z &= 0 \\ 2x - 2y + 2z &= 4 \\ 3x + 9y &= 3 \end{cases}$$

1. $R_2 \rightarrow R_2 - 2R_1$
2. $R_3 \rightarrow R_3 - 3R_1$
3. $R_2 \rightarrow ? \frac{1}{4}R_2$
4. $R_3 \rightarrow R_3 - 6R_2$

$$\begin{cases} x + y + 3z &= 0 \\ y + z &= -1 \\ -15z &= 9 \end{cases}$$

The system of linear equations we started with.

So the original system also has exactly one solution

$$\begin{cases} x = 11/5 \\ y = -2/5 \\ z = -3/5 \end{cases}$$

We used back-substitution to deduce that this system has exactly one

solution $\begin{cases} x = 11/5 \\ y = -2/5 \\ z = -3/5 \end{cases}$

Some language: Row equivalence

Video 2.1.12

Definition

Two matrices or augmented matrices M and M' are said to be **row equivalent** if there is a **sequence** of **elementary row operations** which transforms M into M' .

Example:

In the previous example we saw that the following two augmented matrices are row equivalent:

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 2 & -2 & 2 & 4 \\ 3 & 9 & 0 & 3 \end{array} \right] \xrightarrow{\begin{array}{l} 1. \quad R_2 \rightarrow R_2 - 2R_1 \\ 2. \quad R_3 \rightarrow R_3 - 3R_1 \\ 3. \quad R_2 \rightarrow -\frac{1}{4}R_2 \\ 4. \quad R_3 \rightarrow R_3 - 6R_2 \end{array}} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -15 & 9 \end{array} \right]$$

Some language: Row equivalence

Definition

Two matrices or augmented matrices M and M' are said to be **row equivalent** if there is a **sequence** of elementary row operations which transforms M into M' .

Exercise (on Problem sheets):

→ talking a set and splitting it into different subsets.

Explain why this defines an **equivalence relation** on the set of matrices of a fixed shape.

To prove "is row equivalent" defines an equivalence relation:

- ① Reflexive. $M \sim M \rightarrow$ every matrix is related to itself
- ② Symmetry $\rightarrow$ is equivalent to
If $M \sim M'$ then $M' \sim M$
- ③ Transitivity
If $M \sim N$, $N \sim P$ then $M \sim P$

Using the formulation “row equivalence” we can summarize this strategy for solving systems of linear equations as follows:

Theorem

If two systems of linear equations have row equivalent augmented matrices then their set of solutions are exactly the same.

An illustration of the previous theorem:

